



Transient vibrational power flows in slender random structures: Theoretical modeling and numerical simulations

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ABSTRACT

This paper deals with some recent developments for the modeling and numerical simulation of high-frequency (HF) vibrations of randomly heterogeneous slender structures. A time-domain approach is proposed to take account of the multiple reflection and scattering of vibrational waves in this frequency range. The mathematical model is derived from the semiclassical analysis of strongly oscillating (HF) solutions of quantum or classical wave systems, including acoustic, electromagnetic, or in the present case, elastic waves. This theory shows that the phase space energy density associated to these waves satisfies a radiative transfer equation in a random medium at length scales comparable to the small wavelength. The proposed model also considers energetic boundary and interface conditions consistent with the boundary and interface conditions imposed on the solutions of the underlying wave system. They are given in the form of power flow reflection/transmission operators for the energy rays impinging on a boundary or an interface. Nodal/spectral discontinuous “Galerkin” finite element methods and Monte-Carlo methods are implemented to integrate the radiative transfer equations supplemented with boundary and interface conditions. Some numerical simulations are presented to illustrate the theory: the first one deals with an assembly of random thick beams, and the second one with an assembly of random thick shells. This research applies to the prediction of the linear transient responses of engineering structures to impact loads or shocks, as encountered in the aerospace industry for example.

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1. Introduction

Spatial structures are often subjected to impulse loads which induce high-frequency (HF) vibrations. Here “high” refers to a range of frequencies that extends to many times the fundamental natural frequency of the system of concern, possibly up to the ultrasonic range. Such loads can damage the integrity of the structures or the equipment attached to them. For example, the solar panels unfolding is set off by a pyrotechnic shock leading to the propagation of HF waves in the panels. Another example is the pyrotechnic cut for the space launcher/satellite separation. Thus it is necessary to improve the prediction of the HF structural responses with efficient theoretical and numerical models.

The vibrational response of engineering structures to low-frequency excitations can be predicted efficiently by reduced numerical models of limited size. They are constructed by a Ritz–Galerkin projection on a finite number of eigenmodes associated to the first eigenvalues of the conservative structure. Their efficiency is because, only a low number of generalized degrees of freedom are used in this type of representation.

The normal eigenmodes are obtained by solving a well-posed eigenvalue problem of which only the first eigenvalues and eigenvectors need be computed. These models, eventually updated by appropriate methods [1], compare usually very well with experimental data at low frequencies but may deteriorate rapidly when the frequency is increased. That such techniques are less adapted to the HF range originates from numerical, but above all more fundamental physical reasons. Indeed, the quality of the computed modal reduction basis deteriorates because of the existing numerical algorithms for eigenvalue problems, but also because of the increased influence of the irreducible uncertainties related to the higher order natural modes and frequencies [2]. Observing the dynamic responses of complex structures as the frequency increases, it brings to our notice the fact that the latter have a very different behavior when compared to the ones observed in the low-frequency modal range; see for example the experiments described in [3,4]. This is due to the reflection and scattering of vibrational waves by structural heterogeneities or discontinuities of geometry or material properties, which generally increase as the wavelength decreases (or as the frequency increases). Such a phenomenon contributes to a redistribution of the incident wave energy into many directions. Thus alternative modeling strategies and numerical methods have to be considered for the simulation and prediction of structural responses in the

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HF range; in particular, they should take account of the various sources of uncertainties [5]. Several heuristic strategies have been developed by engineers in this respect. Global approaches like the Statistical Energy Analysis (SEA) [6] of structural-acoustic systems give information on the mean energy level in each substructures. An important difficulty is the estimation of some core parameters such as the coupling loss factors or the power input. Local approaches like the Vibrational Conductivity Analogy (VCA) [7,8] have also been investigated. The point of issue of the existing formulations is their extension to complex engineering structures, as defined in e.g. [5,9].

The latter exhibit typical transport and diffusive behaviors in the HF range of vibration [3,4] as a consequence of the multiple reflection and scattering processes outlined above. This feature has also been identified in the earlier study [5] in terms of apparent induced damping. The transport and diffusive regimes are in this respect relevantly described in the transient domain. As proved in [10], linear transport equations apply to the energy density associated to the strongly oscillating (HF) solutions of the elastic wave equation. More generally this result holds for all symmetric hyperbolic systems, including quantum waves and classical acoustic or electromagnetic waves [11,12]. It has been specialized to slender visco-elastic structures, typically beams, plates and shells, and fluid-saturated poro-visco-elastic media in [13–16] for applications to aerospace structures. The mathematical model is derived from the semiclassical analysis of HF solutions of such wave systems. In addition, the theory shows that the associated energy density in a random medium with correlation lengths comparable to the small wavelength, satisfies a radiative transfer equation (RTE) in the phase space position $\times$ wave vector [12,17–21]. The RTE allows to track the energy paths (rays) within the propagation medium as well. Thus it also describes all power flows in slender structures in the transient domain. The aforementioned analysis provides a rational framework for the validation, and generalization, of SEA or VCA in the time and frequency domains, with due consideration of uncertainties embraced by a random medium model; see for example the discussion in [22].

The main difficulties to date of the proposed approach lie in the consideration of boundary and interface conditions for quadratic quantities, e.g. the energy and power flow densities, consistent with the boundary and interface conditions imposed to the displacement and stress fields of the original elastic wave system. The former are given as power flow reflection/transmission operators for the energy rays impinging on a boundary or an interface. Specular-like transverse boundary reflections, diffuse reflections, or fluid-structure coupling may be treated as a particular case. A formal derivation of these operators at interfaces between slender sub-structures such as beams or shells has been proposed in [16] for the HF range. More recently, a systematic derivation of the elastic energy density boundary conditions near the doubly-hyperbolic set (where both compressional and shear elastic waves are incident on the boundary) and the hyperbolic-elliptic set (where shear waves are incident on the boundary with an angle greater than their critical angle) has been developed in [10]. The analysis however does not account for the glancing set (at critical incidence) and the transport of energy along an interface by so-called gliding rays; this issue is the subject of ongoing research. Yet a radiative transfer model in bounded media with general transverse or diffuse boundary conditions for the power flows may be derived from these results [23].

RTEs can be solved numerically by various methods [24,25], including Monte-Carlo methods [26] or nodal/spectral discontinuous finite element methods [27–31]. Both techniques offer the great advantage of being numerically weakly dissipative and weakly dispersive depending on the order of interpolation. Thus they are

well suited for long-time simulations, when RTEs converge to diffusion equations [12,22,25]. The diffusive regime is characterized by energy equilibration rules which can be exhibited from direct numerical simulations using those schemes [23]. Its onset at late times is prompted by the multiple scattering of energy rays on random heterogeneities in the medium—as modeled by the RTEs themselves—and on boundaries and interfaces. This behavior is in full agreement with the observed main features of high-frequency structural dynamics already sketched above. The diffusive regime is precisely the one invoked in the SEA and VCA approaches to derive steady-state global or local diffusion equation by some heuristic arguments. The main purposes of this paper are twofold. First, the radiative transfer theory actually yielding diffusion for HF structural vibrations is outlined. Second, numerical methods applicable to solve transient RTEs in view of demonstrating their diffusive asymptotics at late times are proposed. It is organized as follows. First, the transport model for the evolution of the high-frequency energy density in randomly heterogeneous slender structures is resumed, considering boundary and interface conditions as well. The following Section 2 also introduces the high-frequency setting for our derivation and a formal justification of the consideration of energy (quadratic) quantities for the description of HF wave propagation in random media. The numerical methods implemented for solving the corresponding RTEs are detailed in Section 3. Then two numerical examples are introduced in Section 4, which resumes and extends the numerical simulation of transient transport in an assembly of random thick beams or shells depicted in [23]. At last Section 5 offers some conclusions and directions of future works.

2. Kinetic model for high-frequency elastic wave propagation in random media

This section first introduces the high-frequency setting applicable to the problems considered in this paper. In Section 2.1 below, the relevance of using quadratic observables and mathematical tools as the Wigner transform to describe high-frequency waves are motivated as well, based on some simple illustrations. Then in Section 2.2 the basic results obtained in [12,17,19] for the transport properties of HF waves in random elastic media are resumed. The boundary conditions to be added to the evolution model recalled below are detailed in Section 2.3 for specular-like or diffusive-like reflections at outer boundaries and in Section 2.4 for reflections and transmissions at interfaces.

2.1. High-frequency setting

Let us consider an heterogeneous elastic body embedded in the open domain $\mathcal{O} \subseteq \mathbb{R}^d$, where $\mathbb{R}$ is the set of all real numbers and $d = 1, 2$ or 3 is the physical space dimension. It is assumed that this medium exhibits (statistically) homogeneous random perturbations of its mechanical parameters, such as the density and the Young or shear modulus. In addition, the scale of variation of these random perturbations, or their correlation length a , is comparable to the wavelength $\lambda \sim a$ considered: this is the so-called *weak coupling* regime [32], where interactions between waves and heterogeneities are expected to be of critical importance. The invoked “weakness” refers to the fact that weak random perturbations are considered, but their overall effect on some macroscopic quantities such as the energy density becomes significant only after a sufficiently long observation time or distance $L \gg \lambda$. These long times/distances are the (macroscopic) regime we are interested in, with the objective of capturing the effect of material randomness at the wavelength scale by means of an effective macroscopic model smoothing out these small scales, normalized as $\varepsilon = \frac{\lambda}{L} \sim \frac{a}{L} \ll 1$. Thus the “weak”

fluctuations considered here are not expected to have a small effect at the macroscopic level. In this setting, the density ϱ^ε and Lamé parameters λ^ε and μ^ε of an isotropic elastic medium are written as

$$\begin{aligned}\varrho^\varepsilon(\mathbf{x}) &= \underline{\varrho}(\mathbf{x}) \left[1 + \sqrt{\varepsilon} X_1 \left(\frac{\mathbf{x}}{\varepsilon} \right) \right], \\ \frac{1}{\lambda^\varepsilon(\mathbf{x})} &= \frac{1}{\underline{\lambda}(\mathbf{x})} \left[1 + \sqrt{\varepsilon} X_2 \left(\frac{\mathbf{x}}{\varepsilon} \right) \right], \\ \frac{1}{\mu^\varepsilon(\mathbf{x})} &= \frac{1}{\underline{\mu}(\mathbf{x})} \left[1 + \sqrt{\varepsilon} X_3 \left(\frac{\mathbf{x}}{\varepsilon} \right) \right],\end{aligned}\quad (1)$$

where $(X_j(\mathbf{x}), \mathbf{x} \in \mathbb{R}^d, j = 1, 2, 3)$, are three real-valued homogeneous and centered, second order stochastic processes. $\underline{\varrho}(\mathbf{x}) = \mathbb{E}\{\varrho^\varepsilon(\mathbf{x})\}$, $\underline{\lambda}(\mathbf{x}) = \mathbb{E}\{\lambda^\varepsilon(\mathbf{x})\}^{-1}$, and $\underline{\mu}(\mathbf{x}) = \mathbb{E}\{\mu^\varepsilon(\mathbf{x})\}^{-1}$ are the averaged macroscopic density and Lamé moduli, respectively, $\mathbb{E}\{X\}$ standing for the mathematical expectation of a random variable X . The perturbations amplitudes in the foregoing model are chosen as the square root of the normalized small wavelength ε in order to avoid localization if they were too large, while this scaling still allows the energy within the medium to be significantly modified by materials fluctuations [12,17,19], as outlined above. Eq. (1) is general enough to embed the occurrence of discrete scatterers, in which case the random processes $(X_j(\mathbf{x}), \mathbf{x} \in \mathbb{R}^d)$ are simply random characteristic functions, as in e.g. [33]. Note that the Lamé parameters depend nonlinearly on Young's modulus and Poisson's ratio, thus random variables X_2 and X_3 are *a priori* correlated. They may be correlated to the density fluctuations X_1 as well. In fact the model of Eq. (1) can support any kind of correlation structures between these perturbations and does not enforce any prior functional form to remain valid. Though particular analytical expressions are considered in the subsequent numerical examples of Section 4, these choices are independent of the theory presented here.

The observables for which limiting models are available in the HF limit $\varepsilon \rightarrow 0$ are not the wave fields u_ε themselves but rather field-field correlations, which are quadratic observables in these wave fields. This is understood considering a strongly oscillating sequence about its zero mean: what can be seen at the macroscopic scale is only that mean, while quadratic observables allow to see the influence of oscillations. Consider a real function $x \mapsto u_\varepsilon(x)$ defined on $\mathbb{R}$ and oscillating with a slowly varying amplitude A , for example:

$$u_\varepsilon(x) = A(x) \sin \frac{x}{\varepsilon}, \quad \varepsilon \ll 1.$$

This function has no strong limit when $\varepsilon \rightarrow 0$, however if its vague limit is of interest, namely:

$$\lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}} \varphi(x) (u_\varepsilon(x))^2 dx,$$

where φ is a continuous function having a compact support in $\mathbb{R}$, then it is found that:

$$\lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}} \varphi(x) (u_\varepsilon(x))^2 dx = \frac{1}{2} \int_{\mathbb{R}} \varphi(x) (A(x))^2 dx$$

as a consequence of Riemann–Lebesgue's lemma.¹ Thus the associated “energy” admits, locally at any point, a limit given by $\frac{1}{2}A(x)^2$; see Fig. 1 below. It allows to “see” the influence of oscillations with amplitude A at any point of interest x selected by the observation function φ , whereas the weak limit of u_ε does not. At last it no longer has the oscillatory feature of the initial function, a simplification which is of high interest for experiments and simulations.

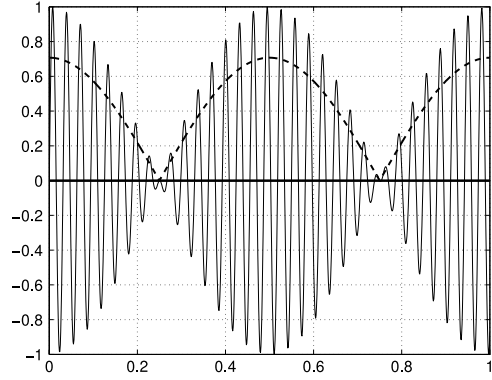


Fig. 1. The function $x \mapsto u_\varepsilon(x) = A(x) \sin \frac{x}{\varepsilon}$ (thin solid line), its mean (thick solid line) and its square root vague limit $x \mapsto 2^{-\frac{1}{2}} A(x)$ (thick dashed line).

Now considering wave propagation in a random medium, multiple scattering by the heterogeneities creates waves $u_\varepsilon(\mathbf{x}, t)$ spreading in many directions at each point in the physical space $\mathcal{O}$ and time $t \in \mathbb{R}$. This motivates the consideration of quadratic observables resolved in the phase space $T^*\mathcal{O} \equiv \mathcal{O} \times \mathbb{R}^d_{\mathbf{k}}$, where $\mathbf{k}$ is the wave vector corresponding to an $\mathbf{x} \rightarrow \mathbf{k}$ Fourier transform of the field-field correlations. A natural mathematical tool describing these correlations is the Wigner transform and its high-frequency limit, the Wigner or semiclassical measure [10,11,32,34], which can be seen as a mathematical generalization of the concepts and ideas introduced just above. Let φ be a smooth, rapidly decreasing function defined on $T^*\mathbb{R}^d$, a so-called observable, and for a square integrable function u defined on $\mathbb{R}^d$ consider the linear operator:

$$\begin{aligned}\Phi(\mathbf{x}, \varepsilon D)u(\mathbf{x}) &= \Phi(\mathbf{x}, \varepsilon D) \left(\int_{\mathbb{R}^d} e^{i\mathbf{k} \cdot \mathbf{x}} \widehat{u}(\mathbf{k}) \frac{d\mathbf{k}}{(2\pi)^d} \right) \\ &= \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{i\mathbf{k} \cdot \mathbf{x}} \varphi(\mathbf{x}, \varepsilon \mathbf{k}) \widehat{u}(\mathbf{k}) d\mathbf{k},\end{aligned}$$

where $\widehat{u}(\mathbf{k}) = \int_{\mathbb{R}^d} e^{-i\mathbf{k} \cdot \mathbf{x}} u(\mathbf{x}) d\mathbf{x}$ is the Fourier transform of $u(\mathbf{x})$. Then for a uniformly bounded sequence (u_ε) in the set of square integrable functions $L^2(\mathbb{R}^d)$ equipped with the scalar product $(u, v)_{L^2} = \int_{\mathbb{R}^d} u(\mathbf{x}) v(\mathbf{x}) d\mathbf{x}$, one has:

$$\lim_{\varepsilon \rightarrow 0} (\Phi(\mathbf{x}, \varepsilon D)u_\varepsilon, u_\varepsilon)_{L^2} = \int_{T^*\mathbb{R}^d} \varphi(\mathbf{x}, \mathbf{k}) W[u_\varepsilon](d\mathbf{x}, d\mathbf{k}), \quad \forall \varphi \quad (2)$$

and the positive, Hermitian measure $W[u_\varepsilon]$ always exists—up to extracting a sub-sequence if need be. It is the so-called Wigner measure of the sequence (u_ε) because it can also be interpreted as the weak limit of its Wigner transform $W_\varepsilon[u_\varepsilon] := W_\varepsilon(u_\varepsilon, u_\varepsilon)$. Indeed, the latter being defined (for temperate distributions u, v) by:

$$W_\varepsilon(u, v)(\mathbf{x}, \mathbf{k}) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{i\mathbf{k} \cdot \mathbf{y}} u(\mathbf{x} - \varepsilon \mathbf{y}) v(\mathbf{x})^* d\mathbf{y}, \quad (3)$$

where $u^* = \bar{u}^\top$ ($\bar{u}$ stands for the complex conjugate and $u^\top$ stands for the transpose of u), one has:

$$(\Phi(\mathbf{x}, \varepsilon D)u, v)_{L^2} = \int_{T^*\mathbb{R}^d} \varphi(\mathbf{x}, \mathbf{k}) W_\varepsilon(u, v)(d\mathbf{x}, d\mathbf{k}).$$

$W[u_\varepsilon]$ describes the limit energy of the sequence (u_ε) , and the function $\varphi(\mathbf{x}, \mathbf{k})$ can be used to select any quadratic observable or quantity of interest associated to this energy: the kinetic energy, or the free energy, or the power flow, etc. For example, the high-frequency ($\varepsilon \rightarrow 0$) strain energy $\mathcal{E}_d(t)$ in $\mathcal{O}$ may be obtained with $\varphi(\mathbf{x}, \mathbf{k}) \equiv \Gamma(\mathbf{x}, \mathbf{k}) = (\underline{\lambda}(\mathbf{x}) + \underline{\mu}(\mathbf{x})) \mathbf{k} \otimes \mathbf{k} + \underline{\mu}(\mathbf{x}) |\mathbf{k}|^2 \mathbf{I}_d$, the acoustic tensor of the unperturbed isotropic medium:

$$\mathcal{E}_d(t) = \frac{1}{2} \text{Tr} \int_{T^*\mathcal{O}} \Gamma(\mathbf{x}, \mathbf{k}) W[u_\varepsilon(\cdot, t)](d\mathbf{x}, d\mathbf{k}).$$

¹ $\int_I f(t) e^{-i\omega t} dt \rightarrow 0$ when $\omega \rightarrow \pm\infty$ for all integrable functions f on an interval I of $\mathbb{R}$.

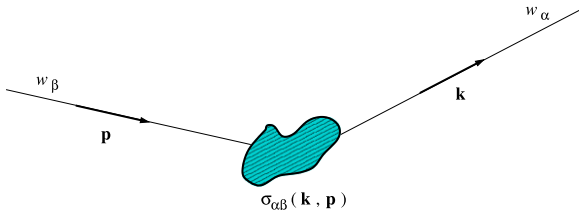


Fig. 2. Scattering cross-section $\sigma_{\alpha\beta}(\mathbf{x}, \mathbf{k}, \mathbf{p})$ in a time-independent randomly heterogeneous medium.

Here $\text{Tr } \mathbf{M}$ stands for the trace of a square matrix $\mathbf{M}$, and $\mathbf{I}_d$ is the $d \times d$ identity matrix. A similar concept to the Wigner measure is H-measure, or microlocal defect measure [35,36]. The main difference is that the latter is associated to any mean-zero square integrable sequence, and it does not require any explicit scale ε as in (3). However it is defined on the unit cosphere bundle $S^*\mathbb{R}^d \equiv \mathbb{R}_x^d \times \mathbb{S}_{\hat{\mathbf{k}}}^{d-1}$, that is it contains slightly less information than the Wigner measure since it does not depend on the norm of the wave vector.

2.2. Radiative transfer of elastic waves in an open domain

Introducing the Wigner measure of the wave field solutions of the elastic wave equation in $\mathcal{O}$, it can be shown along the same lines as above that the high-frequency space–time energy density $\mathcal{E}(\mathbf{x}, t) \in \mathbb{R}_+$ (the sum of the kinetic and strain local energy densities) and local power flow density $\Pi(\mathbf{x}, t) \in \mathbb{R}^d$ associated to these fields may be written [10–12]:

$$\mathcal{E}(\mathbf{x}, t) = \frac{1}{2} \sum_{\alpha=1}^M \int_{\mathbb{R}^d} w_{\alpha}(\mathbf{x}, \mathbf{k}, t) d\mathbf{k}, \quad (4)$$

$$\Pi(\mathbf{x}, t) = \frac{1}{2} \sum_{\alpha=1}^M \int_{\mathbb{R}^d} w_{\alpha}(\mathbf{x}, \mathbf{k}, t) \mathbf{c}_{\alpha}(\mathbf{x}, \mathbf{k}) d\mathbf{k}.$$

M is the number of propagation modes, or polarizations, in the medium, $\{\mathbf{c}_{\alpha}\}_{1 \leq \alpha \leq M}$ are their group velocities, and $\{w_{\alpha}\}_{1 \leq \alpha \leq M}$ are angularly resolved energy densities for these modes living in the phase space $T^*\mathcal{O} \ni (\mathbf{x}, \mathbf{k})$. Typically $M = 3$ and $\alpha = \text{P, SV or SH}$ in an isotropic elastic medium, for the compressional and shear waves respectively. These phase space densities w_{α} , also called hereafter specific intensities, satisfy in addition the multigroup radiative transfer equations (MRTE) [12,17,19]:

$$\begin{aligned} \partial_t w_{\alpha} + \{\Lambda_{\alpha}, w_{\alpha}\} + \Sigma_{\alpha} w_{\alpha} \\ = \sum_{\beta=1}^M \int_{\mathbb{S}^{d-1}} \sigma_{\alpha\beta}(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{p}}) w_{\beta}(\mathbf{x}, \mathbf{p}, t) d\Omega(\hat{\mathbf{p}}), \end{aligned} \quad (5)$$

where $\Lambda_{\alpha}(\mathbf{x}, \mathbf{k}) := c_{\alpha}(\mathbf{x})|\mathbf{k}|$ is the eigenfrequency of the mode α , $\mathbf{c}_{\alpha}(\mathbf{x}, \mathbf{k}) = \nabla_{\mathbf{k}} \Lambda_{\alpha}(\mathbf{x}, \mathbf{k}) = c_{\alpha}(\mathbf{x}) \hat{\mathbf{k}}$, $\hat{\mathbf{k}} := \mathbf{k}/|\mathbf{k}|$ is the unit wave vector in $\mathbb{S}^{d-1}$, the unit sphere of $\mathbb{R}^d$ with the uniform probability measure Ω , and $\{f, g\} = \nabla_{\mathbf{k}} f \cdot \nabla_{\mathbf{x}} g - \nabla_{\mathbf{x}} f \cdot \nabla_{\mathbf{k}} g$ is the usual Poisson bracket. The scattering cross-section $\sigma_{\alpha\beta} \geq 0$ gives the rate of conversion of an energy mode β traveling in the direction $\hat{\mathbf{p}}$ to another mode α in the direction $\hat{\mathbf{k}}$, at position $\mathbf{x}$ and wavenumber $|\mathbf{k}|$, due to scattering on the random inhomogeneities of the medium as described by the model (1); see Fig. 2. At last the total scattering cross-section $\Sigma_{\alpha} \geq 0$ for the polarization α is defined by:

$$\Sigma_{\alpha}(\mathbf{x}, \mathbf{k}) = \sum_{\beta=1}^M \int_{\mathbb{S}^{d-1}} \sigma_{\alpha\beta}(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{p}}) d\Omega(\hat{\mathbf{p}}). \quad (6)$$

In a deterministic medium all scattering cross-sections are zero and right-hand-sides vanish in Eq. (5), which reduces to Liouville

transport equations. Here the kernels $\sigma_{\alpha\beta}$ are assumed to depend on the directions $\hat{\mathbf{k}}$ and $\hat{\mathbf{p}}$ through their scalar product solely, corresponding to isotropic scattering, but more general scattering processes can be considered by which they depend on $\mathbf{k}$ and $\mathbf{p}$. The $\sigma_{\alpha\beta}$'s are given explicitly as functions of the auto- and cross-power spectra of the random perturbations $(X_j(\mathbf{x}), \mathbf{x} \in \mathbb{R}^d)$ of the medium mechanical parameters in Eq. (1) [12,17,19]. In these works, the expressions of the scattering cross-sections are derived from a model of random media exhibiting smooth transitions *a priori* (continuous fluctuations). However their particularization to discrete scatterers is straightforward, using the expression of the scattering cross-section of a single scatterer multiplied by the density of scatterers; see for example [37]. In any case these scattering processes are such that the Hamiltonian Λ_{α} of Eq. (5) is preserved, that is $\Lambda_{\alpha}(\mathbf{x}, \mathbf{k}) = \Lambda_{\beta}(\mathbf{x}, \mathbf{p}) := \omega \in \mathbb{R}$ along the energy paths. These results are adapted to elastic waves in slender structures in [13,15,16], but they also apply to waves in poro-visco-elastic media, acoustic waves, electromagnetic waves, or the Schrödinger equation in random media [14,18,20,21].

For convenience, the MRTE of Eq. (5) are reshaped in the form of a system of first-order conservative equations (8) in the phase space $T^*\mathcal{O}$ as follows. Let us introduce the phase variable $\mathbf{z} = (\mathbf{x}, \mathbf{k}) \in T^*\mathcal{O}$, the set $E = \{1, 2, \dots, M\}$, the vector $\mathbf{w} = (w_1, w_2, \dots, w_M)^T$ of $(\mathbb{R}_+)^M$, and the linear flux operator:

$$\mathcal{F}(\mathbf{w}) = (w_1 \mathbf{F}_1, w_2 \mathbf{F}_2, \dots, w_M \mathbf{F}_M)^T, \quad \mathbf{F}_{\alpha}(\mathbf{z}) = \begin{pmatrix} \mathbf{c}_{\alpha}(\mathbf{z}) \\ -|\mathbf{k}| \nabla_{\mathbf{x}} c_{\alpha} \end{pmatrix}$$

($\mathcal{F}(\mathbf{w})$ is thus a $M \times 2d$ real matrix). The so-called collision operator $\mathcal{Q}$ in Eq. (5) is defined by:

$$\mathcal{Q}(\mathbf{w}) = \mathcal{K}(\mathbf{w}) - \Sigma \mathbf{w}, \quad (7)$$

where $\Sigma(\mathbf{z}) = \text{diag}(\Sigma_{\alpha}(\mathbf{z}))_{\alpha \in E}$, and:

$$\mathcal{K}(\mathbf{w})(\mathbf{z}) = \int_{\mathbb{S}^{d-1}} [\sigma(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{p}})] \mathbf{w}(\mathbf{p}) d\Omega(\hat{\mathbf{p}}),$$

where $[\sigma(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{p}})]_{\alpha\beta} = \sigma_{\alpha\beta}(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{p}})$, $\alpha, \beta \in E$. Then Eq. (5) reads:

$$\partial_t \mathbf{w} + \text{Div}_{\mathbf{z}} \mathcal{F}(\mathbf{w}) = \mathcal{Q}(\mathbf{w}), \quad (8)$$

which is a conservative system in the sense that $\int_{\mathbb{R}^d} \mathcal{Q}(\mathbf{w})(\mathbf{x}, \mathbf{k}) d\mathbf{k} = \mathbf{0}$, $\forall \mathbf{x} \in \mathcal{O}$, whenever $\mathcal{Q}$ is integrable with respect to the phase variable $\mathbf{k} \in \mathbb{R}^d$. On any discontinuity front Σ_D in phase space defined by the hypersurface $\Sigma_D = \{\mathbf{z} \in T^*\mathcal{O}; \mathcal{J}(\mathbf{z}) = 0\}$, its piecewise continuous (weak) solution $\mathbf{w}$ satisfies the Rankine–Hugoniot condition:

$$\sum_{\alpha \in E} \llbracket \Lambda_{\alpha}, \mathcal{J} \rrbracket w_{\alpha} = 0, \quad (9)$$

where $\llbracket f \rrbracket$ stands for the jump of f on the front Σ_D . This relation states that the total normal power flow density $\sum_{\alpha \in E} \{\Lambda_{\alpha}, \mathcal{J}\} w_{\alpha}$ remains continuous across that front; in return the specific intensities w_{α} have no reason to remain continuous themselves. The Rankine–Hugoniot condition (9) does not describe how the individual normal power flows $\{\Lambda_{\alpha}, \mathcal{J}\} w_{\alpha} = (\mathbf{F}_{\alpha} \cdot \hat{\mathbf{n}}_{\Sigma}) w_{\alpha}$ for each mode $\alpha \in E$ are distributed inward and outward on the front of unit outward normal $\hat{\mathbf{n}}_{\Sigma} = \hat{\mathbf{V}}_{\mathbf{z}} \mathcal{J}$. This issue is considered in the next two sections for Σ_D being either a boundary or an interface (a junction) where the group velocities in the medium are discontinuous; in this case $\mathcal{J}$ is *a priori* independent of $\mathbf{k}$.

2.3. Radiative transfer in a bounded domain

The system (8), completed by some initial conditions $\mathbf{w}(\mathbf{z}, 0) = \mathbf{w}^0(\mathbf{z})$, is subsequently considered in the phase space $T^*\mathcal{D} =$

$\mathcal{D} \times \mathbb{R}^d$ where $\mathcal{D}$ is a bounded subset of $\mathcal{O}$, and for $t \in [0, T]$, $T < +\infty$ but arbitrary. It is assumed that $\mathbf{w}^0(\mathbf{x}, \{\mathbf{k} = \mathbf{0}\}) = \mathbf{0}$. Wavenumber $|\mathbf{k}| > 0$ is a fixed parameter, and we shall also assume as usual that the velocities $c_\alpha(\mathbf{x})$ are positive bounded functions (their derivatives $\nabla_{\mathbf{x}} c_\alpha$ are also assumed to be bounded on $\mathcal{D}$) with possible isolated discontinuities if $\mathcal{D}$ contains sharp interfaces between different materials. For any smooth manifold Σ of codimension one in $\mathbb{R}^d$ oriented by its normal $\hat{\mathbf{n}}_\Sigma$, let:

$$\Gamma^\pm(\Sigma) = \{(\mathbf{x}, \mathbf{k}) \in \Sigma \times \mathbb{R}^d; \pm \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_\Sigma(\mathbf{x}) > 0\},$$

$$\Gamma_{\mathbf{x}}^\pm(\Sigma) = \{\mathbf{k} \in \mathbb{R}^d; \pm \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_\Sigma(\mathbf{x}) > 0\}.$$

Then well-posedness of the initial value problem (8) with the aforementioned initial condition is ensured by setting proper boundary conditions on the inflow boundary $\Gamma^-(\partial\mathcal{D})$ of the computational domain $\mathcal{D}$ [38]. Denoting by $\hat{\mathbf{n}}$ the outward unit normal to the (smooth) boundary $\partial\mathcal{D}$, an incoming flux $\mathbf{f}(\mathbf{z}, t)$ can be imposed as $-\mathcal{F}(\mathbf{w})\hat{\mathbf{n}} = \mathbf{f}$ on $\Gamma^-(\partial\mathcal{D})$ and for $t > 0$, where $\mathcal{F}(\mathbf{w})\hat{\mathbf{n}} := ((\mathbf{F}_1 \cdot \hat{\mathbf{n}})w_1, (\mathbf{F}_2 \cdot \hat{\mathbf{n}})w_2, \dots, (\mathbf{F}_M \cdot \hat{\mathbf{n}})w_M)^T$. However generalized boundary reflection laws can also be considered in the form:

$$\mathcal{R}_{\mathbf{x}}(\mathbf{w}) + \mathbf{f} = \mathbf{0} \quad \text{on } \Gamma_{\mathbf{x}}^-(\partial\mathcal{D}), \quad t > 0, \quad (10)$$

where $\mathcal{R}_{\mathbf{x}}$ is a so-called boundary reflection operator, which depends only on the local geometry of the boundary (it may also depend on $|\mathbf{k}|$). This operator describes how the outward normal power flows $\mathcal{F}(\mathbf{w})\hat{\mathbf{n}}, \hat{\mathbf{k}} \cdot \hat{\mathbf{n}} > 0$, are reflected into inward normal power flows $-\mathcal{F}(\mathbf{w})\hat{\mathbf{n}}, \hat{\mathbf{k}} \cdot \hat{\mathbf{n}} < 0$, with possible mode conversions in E . Here the glancing set $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}} = 0$ is ignored, not because such grazing rays are unlikely to occur, but because we lack a theoretical model to describe the associated boundary specific intensities at present. The rigorous construction of $\mathcal{R}_{\mathbf{x}}$ away from the glancing set is detailed in [10] for elastic waves, while a formal derivation for bending, compressional and shear waves in thick beams and shells is given in [16]. The α -th component of this operator has the general form:

$$\mathcal{R}_{\mathbf{x}}(\mathbf{w})_\alpha = \sum_{\beta=1}^M \int_{\hat{\mathbf{p}} \cdot \hat{\mathbf{n}} > 0} \rho_{\alpha\beta}(\mathbf{x}, \hat{\mathbf{k}}, d\hat{\mathbf{p}}) \times (|\mathbf{F}_\beta \cdot \hat{\mathbf{n}}|w_\beta(\mathbf{x}, \mathbf{p}) - |\mathbf{F}_\alpha \cdot \hat{\mathbf{n}}|w_\alpha(\mathbf{x}, \mathbf{k})) \quad \text{on } \Gamma_{\mathbf{x}}^-(\partial\mathcal{D}), \quad (11)$$

for a conservative boundary such that:

$$\sum_{\beta=1}^M \int_{\hat{\mathbf{p}} \cdot \hat{\mathbf{n}} < 0} \rho_{\beta\alpha}(\mathbf{x}, d\hat{\mathbf{p}}, \hat{\mathbf{k}}) = 1 \quad \text{on } \Gamma_{\mathbf{x}}^+(\partial\mathcal{D}),$$

with $\rho_{\beta\alpha}(\mathbf{x}, \hat{\mathbf{p}}, \hat{\mathbf{k}}) = \rho_{\alpha\beta}(\mathbf{x}, -\hat{\mathbf{k}}, -\hat{\mathbf{p}}) = \rho_{\alpha\beta}(\mathbf{x}, \hat{\mathbf{k}}^*, \hat{\mathbf{p}}^*)$, where $\hat{\mathbf{k}}^*(\hat{\mathbf{k}}) := (\mathbf{I}_d - 2\hat{\mathbf{n}} \otimes \hat{\mathbf{n}})\hat{\mathbf{k}}$. Specular reflections for example are $\rho_{\alpha\beta}(\mathbf{x}, \hat{\mathbf{k}}, d\hat{\mathbf{p}}) = \delta_{\alpha\beta} \otimes \delta(\hat{\mathbf{p}} - \hat{\mathbf{k}}^*)$, $\delta_{\alpha\beta}$ standing for the Kronecker symbol; see [23]. These kernels are the power reflection coefficients of the boundary, or reflectances in the optics literature. This boundary scattering process also preserves the Hamiltonian $\omega = \Lambda_\alpha(\mathbf{x}, \mathbf{k}) = \Lambda_\beta(\mathbf{x}, \mathbf{p})$, as the expression of Snell–Descartes law of refraction.

2.4. Radiative transfer with a sharp interface

It is now assumed that the physical domain $\mathcal{D}$ is partitioned into non-overlapping sub-domains and their interface $\Sigma_D \subset \overline{\mathcal{D}}$ is a smooth, bounded manifold of codimension 1 oriented by its unit normal $\hat{\mathbf{n}}_\Sigma$ (ignoring edges and corners at a first glance). More specifically, Σ_D is any interface between these sub-domains or between a sub-domain and the exterior of $\mathcal{D}$ and consequently, it may also be a part or the whole of $\partial\mathcal{D}$. If the group velocities c_α are continuous across Σ_D , the continuity of the normal power flow on

Σ_D is ensured by the conditions $\mathcal{F}(\mathbf{w}^+)\hat{\mathbf{n}}_\Sigma = \mathcal{F}(\mathbf{w}^-)\hat{\mathbf{n}}_\Sigma$ if $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_\Sigma > 0$ (forward flux) and $-\mathcal{F}(\mathbf{w}^-)\hat{\mathbf{n}}_\Sigma = -\mathcal{F}(\mathbf{w}^+)\hat{\mathbf{n}}_\Sigma$ if $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_\Sigma < 0$ (backward flux), where $\mathbf{w}^\pm(\mathbf{x}, \mathbf{k}, t) = \lim_{h \downarrow 0} \mathbf{w}(\mathbf{x} \pm h\hat{\mathbf{n}}_\Sigma, \mathbf{k}, t)$ for a.e. $\mathbf{x} \in \Sigma_D$, and $\mathbf{w}^+(\mathbf{x}, \mathbf{k}, t) = \mathbf{0}$ whenever $\mathbf{x} \in \partial\mathcal{D}$; likewise, $c_\alpha^\pm(\mathbf{x}) = \lim_{h \downarrow 0} c_\alpha(\mathbf{x} \pm h\hat{\mathbf{n}}_\Sigma)$ a.e. $\mathbf{x} \in \Sigma_D$. If however the group velocities are discontinuous across Σ_D , such that $c_\alpha^+ \neq c_\alpha^-$, these expressions may be generalized to the form:

$$\begin{aligned} \mathcal{R}_{\mathbf{x}}^-(\mathbf{w}^-) + \mathcal{T}_{\mathbf{x}}^+(\mathbf{w}^-, \mathbf{w}^+) &= \mathbf{0} \quad \text{on } \Gamma_{\mathbf{x}}^-(\Sigma_D), \\ \mathcal{R}_{\mathbf{x}}^+(\mathbf{w}^+) + \mathcal{T}_{\mathbf{x}}^-(\mathbf{w}^+, \mathbf{w}^-) &= \mathbf{0} \quad \text{on } \Gamma_{\mathbf{x}}^+(\Sigma_D), \end{aligned} \quad (12)$$

$t > 0$, where $\mathcal{R}_{\mathbf{x}}^\pm$ and $\mathcal{T}_{\mathbf{x}}^\mp$ are the upstream (−) and downstream (+) reflection and transmission operators, respectively, of the interface Σ_D . The former have expressions similar to Eq. (11) on either side of Σ_D , while the latter have the form:

$$\begin{aligned} \mathcal{T}_{\mathbf{x}}^+(\mathbf{w}^-, \mathbf{w}^+)_\alpha &= \sum_{\beta=1}^M \int_{\hat{\mathbf{p}} \cdot \hat{\mathbf{n}}_\Sigma < 0} \tau_{\alpha\beta}^+(\mathbf{x}, \hat{\mathbf{k}}, d\hat{\mathbf{p}}) \\ &\times (|\mathbf{F}_\beta \cdot \hat{\mathbf{n}}_\Sigma|w_\beta^+(\mathbf{x}, \mathbf{p}) - |\mathbf{F}_\alpha \cdot \hat{\mathbf{n}}_\Sigma|w_\alpha^-(\mathbf{x}, \mathbf{k})) \quad \text{on } \Gamma_{\mathbf{x}}^-(\Sigma_D), \end{aligned} \quad (13)$$

interchanging the + and − signs for $\mathcal{T}_{\mathbf{x}}^-(\mathbf{w}^+, \mathbf{w}^-)$. Their kernels – the power transmission coefficients of the interface or transmittances – satisfy the symmetries $\tau_{\beta\alpha}^-(\mathbf{x}, \hat{\mathbf{p}}, \hat{\mathbf{k}}) = \tau_{\alpha\beta}^+(\mathbf{x}, -\hat{\mathbf{k}}, -\hat{\mathbf{p}}) = \tau_{\alpha\beta}^+(\mathbf{x}, \hat{\mathbf{k}}^*, \hat{\mathbf{p}}^*)$ for a conservative interface such that:

$$\sum_{\beta=1}^M \left(\int_{\pm \hat{\mathbf{p}} \cdot \hat{\mathbf{n}}_\Sigma > 0} \rho_{\beta\alpha}^\pm(\mathbf{x}, d\hat{\mathbf{p}}, \hat{\mathbf{k}}) + \int_{\pm \hat{\mathbf{p}} \cdot \hat{\mathbf{n}}_\Sigma < 0} \tau_{\beta\alpha}^\pm(\mathbf{x}, d\hat{\mathbf{p}}, \hat{\mathbf{k}}) \right) = 1$$

on $\Gamma_{\mathbf{x}}^\pm(\Sigma_D)$. The relations (12) show how the inward and outward normal power flows are distributed on the interface: the outward flows are the sum of reflected and transmitted flows, taking into account possible mode conversions in E on either side of that interface, while preserving again a constant Hamiltonian $\Lambda_\alpha(\mathbf{x}, \mathbf{k}) = \Lambda_\beta(\mathbf{x}, \mathbf{p})$. The explicit derivation of the reflection and transmission operators for junctions of thick beams or shells is outlined in [16]. Their general properties and some examples are also given in [23].

3. Numerical methods

Numerical methods with low numerical dispersion and dissipation errors are needed to perform long-time simulations of the MRTE (5) in order to reach its diffusion limit. Monte-Carlo methods and discontinuous finite element methods both have these properties. Monte-Carlo methods are easy to implement since they are based on an intuitive interpretation of radiative transfer equations. They also allow to compute local solutions, a decisive advantage over energetic methods (such as finite elements) for large-scale computations. They converge slowly, but it is always possible to control their accuracy. Their main drawback is their lack of versatility for complex geometries as typically encountered in structural dynamics and acoustics. Finite element methods are, on the contrary, much more flexible and can be applied to truly complex geometries. Among these methods, the discontinuous “Galerkin” (DG) finite element method has originally been introduced by Reed and Hill [27] and Lesaint and Raviart [28] to compute neutronic transfers; it is therefore adapted to the integration of radiative transfer equations. In the DG method, boundary fluxes at the edges or faces of the elements maintain the consistency of the numerical solution with the continuous, non-discretized radiative transfer equations, which are otherwise discretized on each element without any continuity relation between elements. This scheme is recalled in Section 3.2 below; the numerical fluxes are constructed in order to get to satisfy the boundary and interface conditions (10) and (12).

3.1. The Monte-Carlo method

The Monte-Carlo method for solving a transport partial differential equation or a radiative transfer integro-differential equation is based on its interpretation as a Fokker–Planck equation for the marginal probability density $\rho = (\rho_\alpha)_{\alpha \in E}$ of an underlying homogeneous Markov process $\{(\mathbf{Z}_t, A_t), t \geq 0\}$ having its values in $T^*\mathbb{R}^d \times E$ [26]. Let $\mathcal{A}$ be the infinitesimal generator of that process, that is:

$$\mathcal{A}\phi = \lim_{h \downarrow 0} \frac{1}{h} (\mathbb{E}_{\mathbf{z}, \alpha} \{\phi(\mathbf{Z}_h, A_h)\} - \phi(\mathbf{z}, \alpha)),$$

$$\forall \phi \in \mathcal{C}_b^1(T^*\mathbb{R}^d \times E),$$

where $\mathbb{E}_{\mathbf{z}, \alpha} \{\phi(\mathbf{Z}_t, A_t)\} := \mathbb{E}\{\phi(\mathbf{Z}_t, A_t) | (\mathbf{Z}_t, A_t)_{t=0} = (\mathbf{z}, \alpha)\}$, and $\mathcal{C}_b^1(T^*\mathbb{R}^d \times E)$ is the set of bounded functions which are $\mathcal{C}^1$ with respect to the physical space variable $\mathbf{x} \in \mathbb{R}^d$. The associated Fokker–Planck, or forward Kolmogorov equation is $\partial_t \rho(\mathbf{z}, t) = \mathcal{A}^* \rho(\mathbf{z}, t)$, where $\mathcal{A}^*$ stands for the adjoint operator. This is exactly the MRTE (8) if $\mathcal{A}$ is chosen as $\mathcal{A}\rho := (\mathbf{F}_\alpha \cdot \nabla_{\mathbf{z}} \rho_\alpha)_{\alpha \in E} + \mathcal{Q}^*(\rho)$, where the adjoint $\mathcal{Q}^*$ of the collision operator (7) is formally constructed by taking the transpose matrix $[\sigma]^T$ as the kernel of $\mathcal{K}^*$. Its solution ρ satisfies:

$$\sum_{\alpha \in E} \int_{T^*\mathbb{R}^d} \phi(\mathbf{z}, \alpha) \rho_\alpha(\mathbf{d}\mathbf{z}, t) = \sum_{\alpha \in E} \int_{T^*\mathbb{R}^d} \mathbb{E}_{\mathbf{z}, \alpha} \{\phi(\mathbf{Z}_t, A_t)\} \rho_\alpha^0(\mathbf{d}\mathbf{z}),$$

$$\forall \phi \in \mathcal{C}_b^1(T^*\mathbb{R}^d \times E),$$

starting from an initial condition $\rho(\mathbf{z}, 0) = \rho^0(\mathbf{z})$. Then taking formally $\phi(\mathbf{z}, \alpha) = \delta(\mathbf{z} - \mathbf{z}_0) \otimes \delta(\alpha - \alpha_0)$ (but $\phi \notin \mathcal{C}_b^1(T^*\mathbb{R}^d \times E)$!) and the usual estimator of the mathematical expectation $\mathbb{E}\{\phi(\mathbf{Z}_t, A_t)\} \simeq \frac{1}{N} \sum_{j=1}^N \phi(\mathbf{z}^j(t), \alpha^j(t))$ constructed by simulating N trajectories $t \mapsto (\mathbf{z}^j(t), \alpha^j(t))$, $1 \leq j \leq N$, of the process $\{(\mathbf{Z}_t, A_t), t \geq 0\}$, yields:

$$w_{\alpha_0}(\mathbf{z}_0, t) \simeq \frac{\varepsilon_0}{N} \sum_{j=1}^N \delta(\mathbf{z}^j(t) - \mathbf{z}_0) \otimes \delta(\alpha^j(t) - \alpha_0),$$

with $\varepsilon_0 = \sum_{\alpha \in E} w_\alpha^0(T^*\mathbb{R}^d)$. Alternatively, if the MRTE (8) is written $\partial_t \mathbf{w} = \mathcal{A} \mathbf{w}$ where $\mathcal{A} \mathbf{w} := -\nabla_{\mathbf{z}} \cdot \mathcal{F}(\mathbf{w}) + \mathcal{Q}(\mathbf{w})$, then its solution in $\mathcal{C}_b^1(T^*\mathbb{R}^d)$ is given by Feynman–Kac’s formula:

$$\mathbf{w}(\mathbf{z}, t) = \mathbb{E}_{\mathbf{z}} \{\mathbf{w}^0(\mathbf{Z}_t)\}. \quad (14)$$

This algorithm extends to a bounded domain $\mathcal{D}$ by constructing the trajectories $t \mapsto (\mathbf{z}^j(t), \alpha^j(t))$ from the broken bicharacteristic flow within $T^*\mathcal{D}$, assuming that its intersection with the glancing set remains empty for all times $t \in [0, T]$. This is practically done by reflecting and transmitting the energy rays in $\mathcal{D}$ according to Snell–Descartes law with all possible mode conversions in E .

3.2. Discontinuous finite element method

3.2.1. Basic principles

Let us now consider the subdivision $\mathcal{T}_h = \cup_{r=1}^{\mathcal{N}} D_r$ of $\mathcal{D}$ into $\mathcal{N}$ non-overlapping sub-domains, or finite elements D_r . A Galerkin variational formulation of the system (8) in an *ad hoc* functional set $\mathcal{W}_h$ defined on $\mathcal{T}_h$ is: Find $\mathbf{w} \in \mathcal{W}_h$ such that

$$\int_{D_r \times \mathbb{R}^d} (\partial_t \mathbf{w} \cdot \mathbf{v} - \mathcal{F}(\mathbf{w}) : \nabla_{\mathbf{z}} \mathbf{v}^T - \mathcal{Q}(\mathbf{w}) \cdot \mathbf{v}) \, \mathbf{d}\mathbf{z}$$

$$+ \int_{\partial D_r \times \mathbb{R}^d} \mathcal{F}(\mathbf{w}) \hat{\mathbf{n}}_r \cdot \mathbf{v} \, \mathbf{d}\gamma(\mathbf{z}) = 0, \quad \forall \mathbf{v} \in \mathcal{W}_h, \quad (15)$$

where $\hat{\mathbf{n}}_r$ is outward unit normal to D_r . If the set $\mathcal{W}_h$ is continuous over adjacent subdomains, summing the foregoing expression yields the standard continuous variational formulation:

$$\int_{T^*\mathcal{D}} (\partial_t \mathbf{w} \cdot \mathbf{v} - \mathcal{F}(\mathbf{w}) : \nabla_{\mathbf{z}} \mathbf{v}^T - \mathcal{Q}(\mathbf{w}) \cdot \mathbf{v}) \, \mathbf{d}\mathbf{z}$$

$$+ \int_{T^*\partial \mathcal{D}} \mathcal{F}(\mathbf{w}) \hat{\mathbf{n}} \cdot \mathbf{v} \, \mathbf{d}\gamma(\mathbf{z}) = 0, \quad \forall \mathbf{v} \in \mathcal{W}_h. \quad (16)$$

If however $\mathcal{W}_h$ is discontinuous over two adjacent sub-domains, then the normal flux $\mathcal{F}(\mathbf{w}) \hat{\mathbf{n}}_r$ in Eq. (15) has to be replaced by a numerical flux $\mathcal{F}_r^*(\mathbf{w}^-, \mathbf{w}^+) \hat{\mathbf{n}}_r$ because the left and right traces of $\mathbf{w}$ on ∂D_r , denoted by $\mathbf{w}^-$ and $\mathbf{w}^+$ respectively, are different *a priori*. The corresponding discontinuous variational formulation is:

$$\int_{D_r \times \mathbb{R}^d} (\partial_t \mathbf{w} \cdot \mathbf{v} - \mathcal{F}(\mathbf{w}) : \nabla_{\mathbf{z}} \mathbf{v}^T - \mathcal{Q}(\mathbf{w}) \cdot \mathbf{v}) \, \mathbf{d}\mathbf{z}$$

$$+ \int_{\partial D_r \times \mathbb{R}^d} \mathcal{F}_r^*(\mathbf{w}^-, \mathbf{w}^+) \hat{\mathbf{n}}_r \cdot \mathbf{v} \, \mathbf{d}\gamma(\mathbf{z}) = 0, \quad \forall \mathbf{v} \in \mathcal{W}_h, \quad (17)$$

with $\mathbf{w}^\pm = \lim_{h \downarrow 0} \mathbf{w}(\mathbf{x} \pm h \hat{\mathbf{n}}_r, \mathbf{k}, t)$ a.e. $\mathbf{x} \in \partial D_r$, and the convention:

$$\mathcal{F}_r^*(\mathbf{w}^-, \mathbf{w}^+) \hat{\mathbf{n}} = \mathcal{F}(\mathbf{w}^-) \hat{\mathbf{n}}, \quad \mathbf{w}^+ = \mathbf{0}, \quad \mathbf{z} \in \Gamma^-(\partial \mathcal{D}).$$

The numerical flux has also to be consistent, that is $\mathcal{F}_r^*(\mathbf{w}, \mathbf{w}) = \mathcal{F}(\mathbf{w})$, and conservative, that is $\mathcal{F}_r^*(\mathbf{w}^-, \mathbf{w}^+) \hat{\mathbf{n}}_r + \mathcal{F}_s^*(\mathbf{w}^-, \mathbf{w}^+) \hat{\mathbf{n}}_s = \mathbf{0}$ if $\partial D_r \cap \partial D_s \neq \emptyset$. The above formulation is the so-called “weak” form of the DG method [31]. The “strong” form corresponding to the original DG method introduced by Lesaint and Raviart [28] is derived by integrating the weak form by parts. Let $\mathcal{L} \mathbf{w} := \partial_t \mathbf{w} + \nabla_{\mathbf{z}} \cdot \mathcal{F}(\mathbf{w})$, then:

$$\int_{D_r \times \mathbb{R}^d} (\mathcal{L} - \mathcal{Q}) \mathbf{w} \cdot \mathbf{v} \, \mathbf{d}\mathbf{z}$$

$$= \int_{\partial D_r \times \mathbb{R}^d} (\mathcal{F}(\mathbf{w}^-) - \mathcal{F}_r^*(\mathbf{w}^-, \mathbf{w}^+)) \hat{\mathbf{n}}_r \cdot \mathbf{v} \, \mathbf{d}\gamma(\mathbf{z}),$$

$$\forall \mathbf{v} \in \mathcal{W}_h. \quad (18)$$

This last form can also be seen as a penalization method applied to the boundary fluxes of the sub-domains (or elements). An alternative form for linear systems is the ultra-weak variational formulation proposed by Després [39]. It consists in choosing $\mathcal{W}_h = \{\mathbf{v}, \mathcal{L}^* \mathbf{v} = \mathbf{0}\}$, where $\mathcal{L}^* \mathbf{w} \equiv -\partial_t \mathbf{w} - (\mathbf{F}_\alpha \cdot \nabla_{\mathbf{z}} w_\alpha)_{\alpha \in E}$ is the formal adjoint of $\mathcal{L}$. Then (18) is written as

$$\partial_t \left(\int_{D_r \times \mathbb{R}^d} \mathbf{w} \cdot \mathbf{v} \, \mathbf{d}\mathbf{z} \right) + \int_{\partial D_r \times \mathbb{R}^d} \mathcal{F}_r^*(\mathbf{w}^-, \mathbf{w}^+) \hat{\mathbf{n}}_r \cdot \mathbf{v} \, \mathbf{d}\gamma(\mathbf{z})$$

$$= \int_{D_r \times \mathbb{R}^d} \mathcal{Q}(\mathbf{w}) \cdot \mathbf{v} \, \mathbf{d}\mathbf{z}, \quad \forall \mathbf{v} \in \mathcal{W}_h. \quad (19)$$

The test functions $\mathbf{v}$ have then to be taken as solutions of an adjoint problem, typically plane waves in piecewise homogeneous media [40]: $\mathbf{v}(\mathbf{z}, t) = \mathbf{V} \phi(\mathbf{k} \cdot \mathbf{x} - \omega t)$ such that $\Gamma(\mathbf{z}) \mathbf{V} = \omega \mathbf{V}$, meaning that ω and $\mathbf{V}$ are the eigenvalues and eigenvectors of the acoustic tensor Γ which is simply $\Gamma(\mathbf{z}) = \text{diag}(\Lambda_\alpha(\mathbf{z}))_{\alpha \in E}$ here.

3.2.2. Numerical implementation

The numerical implementation of the DG method expounded above requires three main ingredients, once the subdivision $\mathcal{T}_h$ has been performed: (i) the definition of numerical fluxes $\mathcal{F}_r^*(\mathbf{w}^-, \mathbf{w}^+)$ on all elements, (ii) the choice of the approximation set $\mathcal{W}_h$, and (iii) the choice of a time-integration scheme. The computational domain in phase-space is reduced to the unit cosphere bundle $S^*\mathcal{D} = \mathcal{D} \times \mathbb{S}^{d-1}$ by considering only normalized wave vectors $\hat{\mathbf{k}}$ defined on the unit sphere $\mathbb{S}^{d-1}$ of $\mathbb{R}^d$. For applications to assemblies of slender structures, $d = 1$ (beams) and $\mathbb{S}^0 =$

$\{-1, +1\}$, or $d = 2$ (shells) and the unit circle $\mathbb{S}^1$ is parametrized by $\theta \in [0, 2\pi[$ such that $\hat{\mathbf{k}} = (\cos \theta, \sin \theta)^\top$. In both cases the meshing of $S^*\mathcal{D}$ reads $\mathcal{T}_h = \bigcup_{r=1}^N \mathbf{X}_r = \bigcup_{r=1}^N \mathbf{D}_r \times \mathbb{S}^{d-1}$, denoting by h the largest diameter of the elements $\mathbf{D}_r$.

Regarding (i), the numerical fluxes ensuring the stability and consistency of the DG method for transport equations have the form [30]:

$$\mathcal{F}_r^*(\mathbf{w}^-, \mathbf{w}^+) = \frac{1}{2} (\mathcal{F}(\mathbf{w}^-) + \mathcal{F}(\mathbf{w}^+)) + \mathbf{C}_r (\mathbf{w}^- - \mathbf{w}^+) \cdot \hat{\mathbf{n}}_r,$$

where $\mathbf{C}_r$ is an $M \times M$ non-negative definite matrix. For example, the (upwind) Roe flux corresponds to $\mathbf{C}_r = \frac{1}{2} \text{diag}(|\mathbf{F}_\alpha|)_{\alpha \in E}$ and the Lax–Friedrichs flux corresponds to $\mathbf{C}_r = \frac{1}{2} \text{diag}(|\mathbf{F}_\alpha|)_{\alpha \in E}$. This expression can be generalized as follows for an element boundary $\partial \mathbf{D}_r$ belonging to an interface (a junction) Σ_D where the group velocities c_α are discontinuous. Let $\Sigma_D^r = \partial \mathbf{D}_r \cap \Sigma_D \neq \emptyset$, then having in mind Eq. (12), the (upwind) numerical flux is written as

$$\mathcal{F}_r^*(\mathbf{w}^-, \mathbf{w}^+) \hat{\mathbf{n}}_r = \mathcal{F}(\mathbf{w}^-) \hat{\mathbf{n}}_r \quad (20)$$

on $\Gamma_x^+(\Sigma_D^r)$, and:

$$\mathcal{F}_r^*(\mathbf{w}^-, \mathbf{w}^+) \hat{\mathbf{n}}_r = \mathcal{F}(\mathbf{w}^-) \hat{\mathbf{n}}_r - (\mathcal{R}_x^-(\mathbf{w}^-) + \mathcal{T}_x^+(\mathbf{w}^-, \mathbf{w}^+)) \quad (21)$$

on $\Gamma_x^-(\Sigma_D^r)$. This definition extends in fact to the case $\Sigma_D^r = \emptyset$ if the velocities are continuous across $\partial \mathbf{D}_r$. Then $\mathcal{R}_x^-(\mathbf{w}^-) \equiv \mathbf{0}$ and $\mathcal{T}_x^+(\mathbf{w}^-, \mathbf{w}^+) = (\mathcal{F}(\mathbf{w}^-) - \mathcal{F}(\mathbf{w}^+)) \hat{\mathbf{n}}_r$ on $\Gamma_x^-(\partial \mathbf{D}_r)$, such that $\mathcal{F}_r^*(\mathbf{w}^-, \mathbf{w}^+) = \mathcal{F}(\mathbf{w}^-)$ on $\Gamma_x^+(\partial \mathbf{D}_r)$ and $\mathcal{F}_r^*(\mathbf{w}^-, \mathbf{w}^+) = \mathcal{F}(\mathbf{w}^+)$ on $\Gamma_x^-(\partial \mathbf{D}_r)$.

Regarding (ii), the approximation set $\mathcal{W}_h \subset L^2(S^*\mathcal{D})$ is constructed by tensoring the physical space and phase variables $\mathcal{W}_h = \mathcal{D}_h \otimes \mathcal{S}_h$, with:

$$\mathcal{D}_h = \{\mathbf{v} \in (L^2(\mathcal{D}))^M; v_\alpha|_{\mathbf{D}_r} \in \mathbb{P}_p(\mathbf{D}_r), 1 \leq r \leq N, \alpha \in E\}$$

where $\mathbb{P}_p(\mathbf{D}_r) \subset H^1(\mathbf{D}_r)$ is typically a polynomial set, and $\mathcal{S}_h$ is a finite dimensional set of real-valued functions defined on $\mathbb{S}^{d-1}$. Practically in dimension 1, $\mathbb{P}_p(\mathbf{D}_r)$ is spanned by either a set of Jacobi orthogonal polynomials up to the order p (the modal, or spectral approach), or a set of Lagrange interpolation polynomials associated to $p + 1$ interpolation points (the nodal approach). The degrees of freedom in physical space are the projection coefficients on the Jacobi polynomials of the approximate solution in $\mathcal{W}_h$ in the modal approach, and its values at the interpolation points in the nodal approach. It should be noted that if the latter are chosen as Gauss–Jacobi quadrature points on $[-1, 1]$, the corresponding Lagrange polynomials are computed explicitly as functions of Jacobi polynomials, which yields a unified framework for the modal and nodal approaches. The numerical dispersion and dissipation errors introduced by this spatial discretization are $O(K^{2p+3})$ and $O(K^{2p+2})$, respectively, for decentered numerical fluxes ($\mathbf{C}_r \neq \mathbf{0}$), or $O(K^{\lfloor \frac{p}{2} \rfloor + 3})$ and 0 for centered fluxes ($\mathbf{C}_r = \mathbf{0}$) [41], where $K = h|\mathbf{k}|$ and $\lfloor \cdot \rfloor$ stands for the integer part. These so-called “superconvergence” properties of DG methods are particularly attractive for long-times numerical simulations. The spatial discretization in dimension 2 is achieved by tensoring both dimensions on structured meshes, as proposed in [42] for example. The construction of $\mathcal{S}_h$ is however more involved. The modal approach corresponds to a truncated Fourier expansion in θ , but this scheme does not ensure the positivity of the approximate solution because of the slowness of the convergence of Fourier series. The nodal approach by Lagrange interpolation yields the well-known ray effect which also arises in the discrete ordinate methods widely used in neutronic or thermal transfer computations [24, Chapter 8]. Therefore further researches are needed on this issue; it is the subject of ongoing investigations.

At last, regarding (iii), strong stability-preserving (SSP), high-order Runge–Kutta schemes have been implemented [43]. The CFL condition of these schemes, coupled to the aforementioned phase-space discretization methods, can be computed numerically for various orders p .

4. Numerical examples

In the examples below the radiative transfer equations (5) with specular-like boundary and interface (at the junctions) conditions are solved numerically by the schemes expounded in the foregoing section. In Section 4.1 the example of [16] of scalar transport in coupled Timoshenko beams is resumed and extended to randomly heterogeneous beams. Section 4.2 deals with two coupled thick cylindrical shells.

4.1. Coupled beams

We first consider an assembly of two random Timoshenko beams $] -L, 0[\cup]0, L[$, $L > 0$, with a junction at $x = 0$ such that beam #2 forms an angle ϕ with beam #1. Three types of energy rays propagate in such a system: a pure shear transverse mode at the velocities c_{Tj} ($j = 1$ or 2 for either beam #1 or beam #2) denoted by subscript T, a pure bending mode at the velocities c_{Pj} denoted by subscript Pb, and a longitudinal mode at the velocities c_{Pj} denoted by subscript Pn; see [15,22] for some details on the derivation. Therefore the energy density evolution in this system is depicted by three coupled, one-dimensional radiative transfer equations ($M = 3$ and $d = 1$ such that $\hat{k} \equiv \text{sign}(k)$) for $\alpha \in E = \{T, \text{Pb}, \text{Pn}\}$ written as:

$$\partial_t w_\alpha + c_\alpha \hat{k} \partial_x w_\alpha + (1 + \eta) \Sigma_\alpha(k) w_\alpha = \Sigma_\alpha(k) w_\alpha(-k), \quad (22)$$

where the coupling is ensured by the left (−) and right (+) power reflection and transmission operators $\mathcal{R}_0^\pm$, $\mathcal{T}_0^\pm$ of the junction. Indeed, for thick beams no coupling of modes occurs through the collision operator on the right-hand-side in Eq. (22) [22]. The expressions of the kernels $\rho_{\alpha\beta}^\pm(0, \hat{k}, \hat{p})$ and $\tau_{\alpha\beta}^\pm(0, \hat{k}, \hat{p})$ of $\mathcal{R}_0^\pm$ and $\mathcal{T}_0^\pm$, respectively, as functions of ϕ and beams mechanical and geometrical characteristics are derived in [16], using a high-frequency model compatible with the propagation characteristics obtained by the transport theory. This study has also shown that the power reflection coefficients at a beam end for either Dirichlet (clamped) or Neumann (free) boundary condition were all equal to 1, without mode conversion of any type; therefore specular reflections are considered at the boundaries $x = \pm L$. In agreement with their analytical expressions derived in [15,22], the total scattering cross-sections are written $\Sigma_\alpha(k) = \frac{\pi}{2} \zeta_\alpha c_\alpha k^2 S(2k)$, where the phase function $S(k)$ is the one-dimensional Fourier transform of the normalized correlation function (NCF) $R(x - y)$ of random fluctuations of the density and inverse Young and shear moduli, $S(k) = \frac{1}{2\pi} \int_{\mathbb{R}} R(r) e^{ikr} dr$; and ζ_α is a dimensionless strength factor related to their variation coefficients (or amplitudes). Its analytical expressions for various NCF models are gathered in Table 1 below, where the NCF $R(r)$ is normalized such that $a := \int_0^{+\infty} R(r) dr$ is a correlation length of these random fluctuations. All computations are performed for the initial conditions:

$$w_T^0(x, k) = \varphi\left(\frac{x - x_0}{r_0}\right) \otimes \delta(\hat{k} - \hat{k}_0), \quad w_{\text{Pb}}^0 = w_{\text{Pn}}^0 \equiv 0,$$

with $\varphi(x) = \max(0, 1 - 2|x|)$ (“hat” function). The parameters are chosen as $x_0 = -0.1 \times L$, $r_0 = 0.12 \times L$, and $\hat{k}_0 = +1$. Beam velocities are such that $c_{T2}/c_{T1} = c_{P2}/c_{P1} = \sqrt{2}$, their Poisson’s ratio being $\nu_1 = \nu_2 = 0.3$, and the junction angle is $\phi = \pi/3$.

The coupled radiative transfer equations (22) are solved numerically by the Runge–Kutta DG finite element method expounded in Section 3.2 above, selecting the weak form (17). The assembly is partitioned into 40 uniform spatial elements. Legendre polynomials up to the fifth order are used as local basis functions on them, and Eq. (22) are solved in time by a fourth-order SSP Runge–Kutta scheme [44]. The Courant number for the

Table 1

Phase function models for beams; K_ν is the modified Bessel function of the second kind, Γ is the Gamma function, $\nu \in [0, 1]$ is the Hurst number in von Karman model (note that the latter reduces to Markov model for $\nu = \frac{1}{2}$).

NCF model	$R(r)$	$S(k)$
Gaussian	$\exp\left(-\pi \frac{r^2}{4a^2}\right)$	$\frac{a}{\pi} \exp\left(-\frac{a^2 k^2}{\pi}\right)$
Markov	$\exp\left(-\frac{ r }{a}\right)$	$\frac{a}{\pi} (1 + a^2 k^2)^{-1}$
von Karman	$\frac{2^{1-\nu}}{\sqrt{\pi} \Gamma(\nu + \frac{1}{2})} \left(\frac{ r }{a}\right)^\nu K_\nu\left(\frac{ r }{a}\right)$	$\frac{a}{\pi} (1 + a^2 k^2)^{-(\nu + \frac{1}{2})}$
Rayleigh	$\delta\left(\frac{ r }{2a}\right)$	$\frac{a}{\pi}$

simulations is fixed at $c \equiv c_{T2} \frac{\Delta t}{\Delta x} = 2.5 \cdot 10^{-2}$. Figs. 3 through 5 display our results for $\zeta_\alpha = 0.3$, $ak = 2$, and Gaussian, Markov and Rayleigh models of correlation for the scattering cross-sections, respectively. The first case corresponds to smooth, differentiable random heterogeneities (in the mean-square sense), while the second and third ones correspond to non-differentiable heterogeneities. The time scale is $T = L/c_{T2}$, the time needed by the wave front to reach the extremity $x = L$ of the homogeneous assembly starting from the junction. The thin vertical line indicates the location of the junction. The wavefronts of shear ($\alpha = T$) and bending/axial ($\alpha = P$) energy densities propagating at the speeds c_{Tj} and c_{Pj} , respectively, are marked by small capital letters. A small amount of damping $\eta = 0.01$ has been added such that the total energy within the system at a fixed wavenumber k evolves as (see Eq. (4)):

$$\mathcal{E}(t) = \sum_{\alpha \in E} e^{-\eta \Sigma_\alpha t} \mathcal{E}_\alpha^0, \quad \mathcal{E}_\alpha^0 = \int_{-L}^L w_\alpha^0(x, \pm|k|) dx. \quad (23)$$

This exact result is recovered numerically by our scheme, as shown by Figs. 6 through 8. They display the overall mechanical energies $\mathcal{E}_j^h(t)$ computed from the numerical solutions $w_\alpha^h(x, k, t)$ of (22) within each beam by:

$$\mathcal{E}_1^h(t) = \sum_{\alpha \in E} \int_{-L}^0 w_\alpha^h(x, \pm 1, t) dx,$$

$$\mathcal{E}_2^h(t) = \sum_{\alpha \in E} \int_0^L w_\alpha^h(x, \pm 1, t) dx,$$

and their sum, normalized by the total energy introduced into the system $\mathcal{E}(0) = \sum_\alpha \mathcal{E}_\alpha^0 = \mathcal{E}_T^0$. The computed and exact total energies perfectly match on these plots: one concludes that the proposed scheme in dimension one is numerically conservative.

These results exhibit the following features: one observes a beating phenomenon of the total energy within each subsystem, by which it is transferred back and forth from the first beam to the second one and the other way round. At late time an equipartition regime holds between both beams. It is described by diffusion equations as explained in [22] for example. It should be noted that this diffusive regime also arises in the case of a piecewise homogeneous assembly, the onset of diffusion being prompted by the mixing of wave fronts and modes at the junction, and to a lesser extent at the boundaries. This mixing process is only accelerated by the presence of random heterogeneities, as outlined elsewhere [23]; also the rougher they are, the faster diffusion arises, and backscattering increases. Finally, the long-time limit of the ratio $\mathcal{E}_{Pj}^h(t)/\mathcal{E}_{Tj}^h(t)$ of the bending/longitudinal energy over the shear energy within each beam is proved to be the ratio c_{Tj}/c_{Pj} to the power of the physical dimension, $d = 1$ in the present case, independently of the source type and scatterers [12,22]. Here this universal equilibration rule is recovered by the DG scheme, as exemplified by Fig. 9 for a Gaussian model of correlation. The other NCF models yield the same conclusions (plots not displayed here).

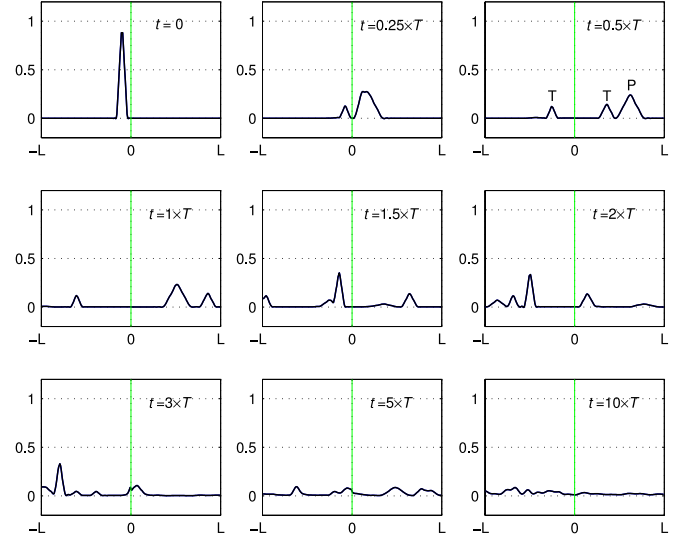


Fig. 3. Evolution of the energy density in an assembly of two randomly heterogeneous Timoshenko beams with $\phi = \frac{\pi}{3}$, $c_{T2}/c_{T1} = c_{P2}/c_{P1} = \sqrt{2}$, $\nu_1 = \nu_2 = 0.3$, $\eta = 0.01$, $T = L/c_{T2}$, and $\zeta_\alpha = 0.3$, $ak = 2$ for the scattering cross-sections with a Gaussian model of correlation.

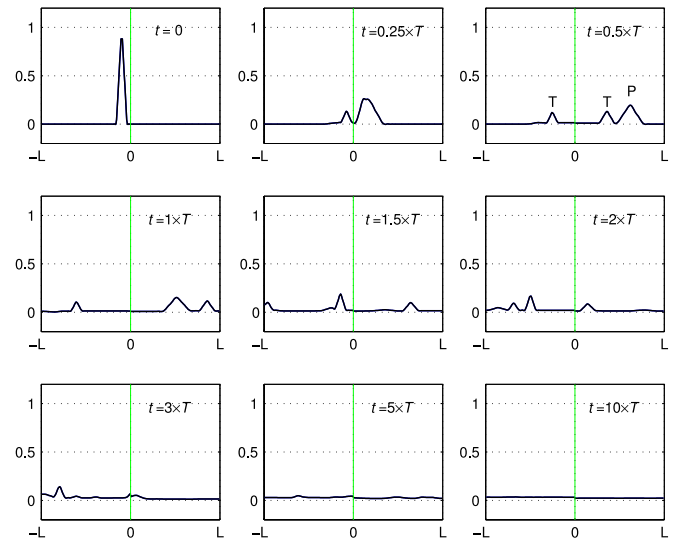


Fig. 4. Same as Fig. 3, with a Markov model of correlation.

4.2. Coupled shells

Next we consider an assembly of two thick cylindrical shells, the first one with a constant radius (shell #1) and the second one with a variable radius (shell #2). Both cylinders have the same length L which is also twice the constant radius of the first shell. The angle between the shells is again denoted by ϕ , and their mean surfaces are denoted by $\mathcal{D}_1$ and $\mathcal{D}_2$, respectively. The junction line is the circle $\Sigma_D = \{\mathbf{x} = (0, y), 0 \leq y \leq \pi L\}$ in curvilinear coordinates. Five types of energy rays propagate in such a system: a pure shear transverse mode at the velocities c_{Tj} ($j = 1$ or 2 for either shell #1 or shell #2) denoted by subscript T, and coupled longitudinal and transverse modes P and S each of multiplicity 2 corresponding to either in-plane (subscript n) or bending energies (subscript b), at the velocities c_{Pj} and c_{Sj} respectively; see [13] for some details on the derivation. Then the energy density evolution in this system is depicted by five coupled radiative transfer equations (5) with $M = 5$ – since $\alpha = T, Pb, Pn, Sb$ or Sn – and $d = 2$. Their coupling is ensured by the left (–) and right (+) power reflection and transmission

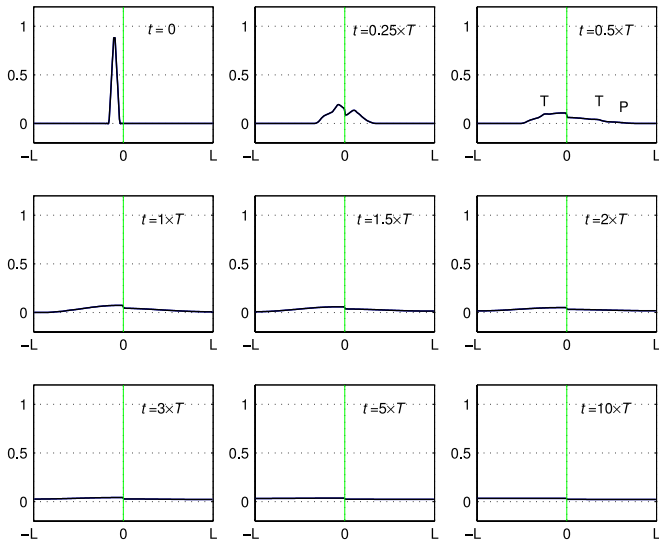


Fig. 5. Same as Fig. 3, with a Rayleigh model of correlation.

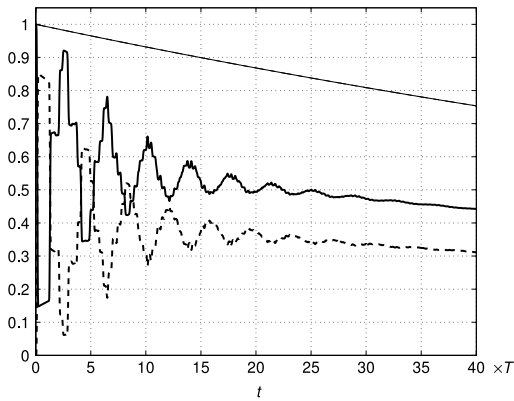


Fig. 6. Evolution of the total vibrational energies in an assembly of two randomly heterogeneous Timoshenko beams with $\phi = \frac{\pi}{3}$, $c_{T2}/c_{T1} = c_{P2}/c_{P1} = \sqrt{2}$, $\nu_1 = \nu_2 = 0.3$, $\eta = 0.01$, $T = L/c_{T2}$, and $\zeta_\alpha = 0.3$, $ak = 2$ for the scattering cross-sections with a Gaussian model of correlation. Thick solid line: beam #1, thick dashed line: beam #2, thin solid line: computed total energy, thin dashed line: exact total energy.

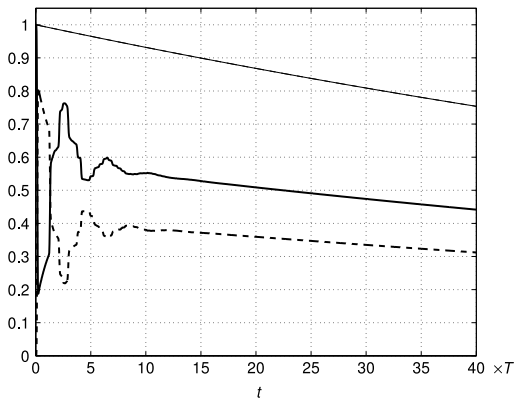


Fig. 7. Same as Fig. 6, with a Markov model of correlation.

operators $\mathcal{R}_y^\pm$, $\mathcal{T}_y^\pm$ of the shell junction with $0 \leq y \leq \pi L$. Their expressions as functions of ϕ , the incident tangent wave vector $\mathbf{k}' = (\mathbf{I}_2 - \hat{\mathbf{n}}(y) \otimes \hat{\mathbf{n}}(y))\mathbf{k}$ on the junction line $\Sigma_D \ni (0, y)$, and shells mechanical and geometrical characteristics are derived in [16], using a high-frequency model compatible with the propagation characteristics obtained by the transport theory within plates.

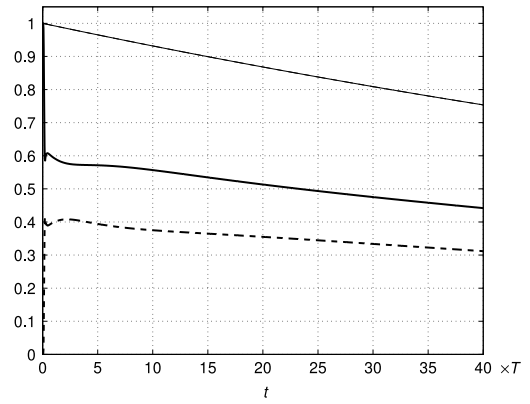


Fig. 8. Same as Fig. 6, with a Rayleigh model of correlation.

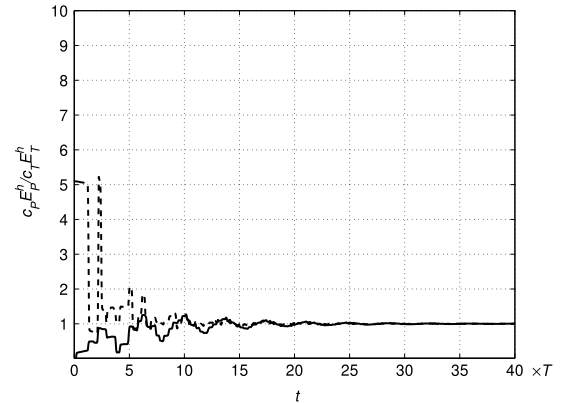


Fig. 9. Evolution of the ratio $c_{Pj}\mathcal{E}_{Pj}^h(t)/c_{Tj}\mathcal{E}_{Tj}^h(t)$ of the total bending/longitudinal and transverse energies in an assembly of two randomly heterogeneous Timoshenko beams with $\phi = \frac{\pi}{3}$, $c_{T2}/c_{T1} = c_{P2}/c_{P1} = \sqrt{2}$, $\nu_1 = \nu_2 = 0.3$, $\eta = 0.01$, $T = L/c_{T2}$, and $\zeta_\alpha = 0.3$, $ak = 2$ for the scattering cross-sections with a Gaussian model of correlation. Solid line: beam #1 ($j = 1$), dashed line: beam #2 ($j = 2$).

As the curvature tensor has no influence on the high-frequency regime in this model [13], the same reflection/transmission kernels as for flat plates are used in this example. Neumann boundary conditions are considered at the extremities of the shells, and the corresponding power reflection coefficients have been derived in [16] as well. Here it has been shown that the shear mode is uncoupled to the in-plane and bending modes by boundary reflections, however the longitudinal P and transverse S modes are coupled. Otherwise all these modes are fully coupled by the junction. The phase function $S(\mathbf{k})$ for the scattering cross-sections derived in [13,22] for shells is now the two-dimensional Fourier transform of the NCF $R(\mathbf{x} - \mathbf{y})$ of random fluctuations of the density and inverse Lamé moduli, $S(\mathbf{k}) = \frac{1}{4\pi^2} \int_{\mathbb{R}^2} R(\mathbf{r}) e^{i\mathbf{k} \cdot \mathbf{r}} d\mathbf{r}$. Its analytical expressions for various NCF models are gathered in Table 2 below, where the correlation function $R(\mathbf{r})$ is normalized such that $\pi a^2 := \int_0^{+\infty} \int_{\mathbb{S}^1} R(r\hat{\mathbf{r}}) r dr d\Omega(\hat{\mathbf{r}})$. The total scattering cross-sections (6) have the form:

$$\Sigma_\alpha(\mathbf{k}) = \frac{\pi}{2} c_\alpha |\mathbf{k}|^3 \sum_\beta \int_{\mathbb{S}^1} f_{\alpha\beta}(\hat{\mathbf{k}} \cdot \hat{\mathbf{p}}) S\left(|\mathbf{k}| \left(\hat{\mathbf{k}} - \frac{c_\alpha}{c_\beta} \hat{\mathbf{p}}\right)\right) d\Omega(\hat{\mathbf{p}}),$$

where $f_{\alpha\beta}$ is a polynomial function, of which expression is derived from the formulas of the scattering cross-sections obtained in [13,22]:

$$\begin{aligned} f_{TT}(x) &= \zeta_{11} + 2\Re\{\zeta_{13}\}x + \zeta_{33}x^2, \\ f_{SS}(x) &= \zeta_{11}x^2 + \zeta_{33}(2x^2 - 1) + 2\Re\{\zeta_{13}\}x(2x^2 - 1), \\ f_{PS}(x) &= (1 - x^2) \left[\zeta_{11} + 2\sqrt{1 - \nu}\Re\{\zeta_{13}\}x + 2(1 - \nu)\zeta_{33}x^2 \right], \end{aligned}$$

Table 2

Phase function models for shells (note that von Karman model reduces to Markov model for the Hurst number $\nu = \frac{1}{2}$).

NCF model	$R(\mathbf{r})$	$S(\mathbf{k})$
Gaussian	$\exp\left(-\frac{ \mathbf{r} ^2}{a^2}\right)$	$\frac{a^2}{4\pi} \exp\left(-\frac{a^2 \mathbf{k} ^2}{4}\right)$
Markov	$\exp\left(-\sqrt{2}\frac{ \mathbf{r} }{a}\right)$	$\frac{a^2}{4\pi} \left(1 + \frac{1}{2}a^2 \mathbf{k} ^2\right)^{-\frac{3}{2}}$
von Karman	$\frac{2^{-\nu}}{\Gamma(\nu+1)} \left(\frac{\sqrt{2} \mathbf{r} }{a}\right)^\nu K_\nu\left(\frac{\sqrt{2} \mathbf{r} }{a}\right)$	$\frac{a^2}{4\pi} \left(1 + \frac{1}{2}a^2 \mathbf{k} ^2\right)^{-\nu-1}$
Rayleigh	$\pi \delta\left(\sqrt{2}\frac{ \mathbf{r} }{a}\right)$	$\frac{a^2}{4\pi}$

and:

$$f_{PP}(\mathbf{x}) = \nu^2 \zeta_{22} + 2\nu \Re\{ \zeta_{12} \} x + (\zeta_{11} + 2\nu(1-\nu) \Re\{ \zeta_{23} \}) x^2 + 2(1-\nu) \Re\{ \zeta_{13} \} x^3 + (1-\nu)^2 \zeta_{33} x^4.$$

The ζ_{jj} 's are the variances of the processes $(X_j(\mathbf{x}), \mathbf{x} \in \mathbb{R}^2)$ in Eq. (1), and the ζ_{jk} 's, $j \neq k$, are coherency coefficients. The initial condition is an isotropic shear impact $w_T^0(\mathbf{x}, \mathbf{k}) = \delta(\mathbf{x} - \mathbf{x}_0)$ independently of $\mathbf{k}$, all other modes being initially unloaded; here $\mathbf{x}_0 = (-L/2, 0)$ in curvilinear coordinates. The Young's moduli, densities, and thicknesses of both shells are equal, such that [13]:

$$\frac{c_{P2}}{c_{P1}} = \sqrt{\frac{1-\nu_1^2}{1-\nu_2^2}}, \quad \frac{c_{S2}}{c_{S1}} = \sqrt{\frac{1+\nu_1}{1+\nu_2}}, \quad \frac{c_{T2}}{c_{T1}} = \sqrt{\frac{6+5\nu_1}{6+5\nu_2}};$$

their Poisson's ratios are $\nu_1 = 0.3$ and $\nu_2 = 0.2$, and the junction angle is $\phi = \pi/12$.

The coupled radiative transfer equations (5) are solved in this case by a direct Monte-Carlo method [26]. The computer code developed for this application has been fully vectorized, in such a way that numerical simulations of $O(10^6)$ ray paths with $O(10^2)$ time steps can be performed in less than one hour of CPU time on a personal computer. Figs. 10 through 16 below display our results for $a|\mathbf{k}| = 2$, perturbation strengths $\zeta_{11} = \zeta_{22} = 0.1$, $\zeta_{33} = 0.9$, $\zeta_{12} = \zeta_{13} = 0$, and $\zeta_{23} = 0.9$ with a Gaussian or Rayleigh model of correlation in shell #1, and no perturbation $\zeta_{jk} = 0$ in shell #2; no damping is accounted for either in both shells. The time scale used in these figures is $T = L/c_{T2}$. Figs. 10 and 11 first plot the scattering cross-sections of shell #1 for this set of parameters. They show that for Gaussian correlations of the Lamé parameters and density, these kernels are rather forward peaked while mode conversion is less important. On the contrary, Rayleigh correlations tend to mix significantly more the scattered directions and modes. Figs. 12 and 13 display the evolution of the energy density within the shell assembly for the Gaussian and Rayleigh models of correlation, respectively, while Figs. 14 and 15 display the evolution of the overall mechanical energies within each shell and their sum. They show that the Monte-Carlo scheme is numerically conservative. In addition Fig. 16 displays the ratios $\mathcal{E}_{pj}^h(t)/\mathcal{E}_{Sj}^h(t)$ of the total bending/longitudinal and transverse energies within each shell, respectively, where:

$$\mathcal{E}_{pj}^h(t) = \sum_{\alpha=\text{Pn,Pb}} \int_{\mathcal{D}_j \times \mathbb{S}^1} w_\alpha^h(\mathbf{x}, \hat{\mathbf{k}}, t) d\mathbf{x} d\hat{\mathbf{k}}, \quad j = 1, 2,$$

and a similar expression for $\mathcal{E}_{Sj}^h(t)$; $w_\alpha^h(\mathbf{x}, \hat{\mathbf{k}}, t)$ being the numerical solutions of (5). All computations exhibit the same features as for the assembly of beams, calling for similar qualitative comments. For an isotropic collision operator, the diffusive limit at large times of $\mathcal{E}_{pj}^h(t)/\mathcal{E}_{Sj}^h(t)$ is proved again to be the ratio c_{Sj}/c_{Pj} to the power of the physical dimension $d = 2$ independently of the source type and scatterers [12,22]. This universal equilibration rule is recovered by the Monte-Carlo scheme as well, see Fig. 16. The Markov and von Karman NCF models yield the same quantitative and qualitative conclusions (plots not displayed here). The outbreak of energy equipartitioning is accelerated for the Rayleigh model

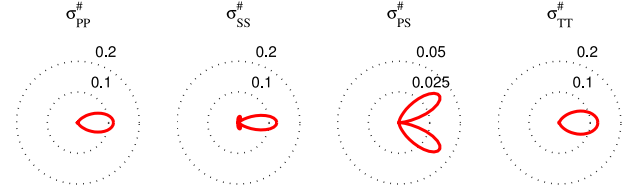


Fig. 10. Normalized scattering cross-sections $\sigma_{\alpha\beta}^{\#} = \sigma_{\alpha\beta}/(\omega a)^2$ of shell #1 as functions of $\Theta = \arccos(\hat{\mathbf{k}} \cdot \hat{\mathbf{p}})$ for $\nu_1 = 0.3$, $a|\mathbf{k}| = 2$ and a Gaussian model of correlation with $\zeta_{11} = \zeta_{22} = 0.1$, $\zeta_{33} = 0.9$, $\zeta_{12} = \zeta_{13} = 0$, $\zeta_{23} = 0.9$.

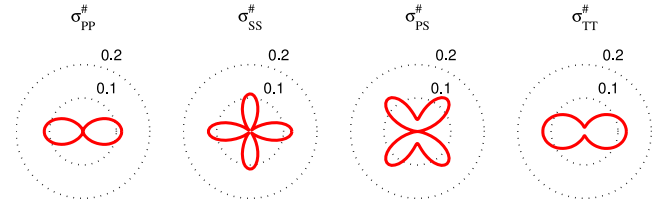


Fig. 11. Same as Fig. 10, for a Rayleigh model of correlation.

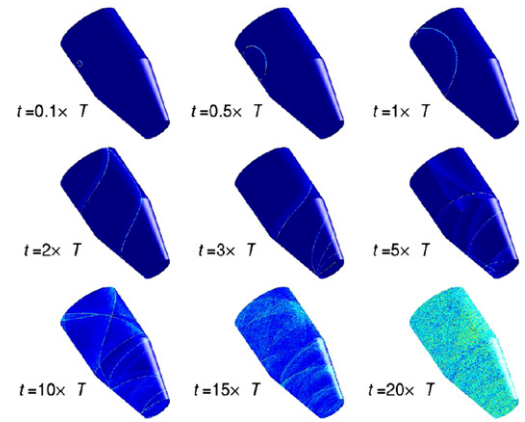


Fig. 12. Evolution of the energy density in an assembly of two randomly heterogeneous cylindrical shells with $\phi = \frac{\pi}{12}$, $\nu_1 = 0.3$, $\nu_2 = 0.2$, $T = L/c_{T2}$, and $a|\mathbf{k}| = 2$, $\zeta_{11} = \zeta_{22} = 0.1$, $\zeta_{33} = 0.9$, $\zeta_{12} = \zeta_{13} = 0$, $\zeta_{23} = 0.9$ for the scattering cross-sections with a Gaussian model of correlation in shell #1.

of correlation, as expected from the shape of the scattering cross-sections which contribute to mix the energy paths more promptly (recall Fig. 11). This conclusion holds for the previous example – beam junction – as well. It should be reminded, however, that these results ignore the contribution of the glancing rays possibly trapped by the boundaries and the junction line, because a theoretical model lacks at present to describe them. Indeed it is known from experiments that the vibrational energy is very much likely to be guided by the interfaces, stiffeners or junctions in a real-life complex structure, so these effects should not be disregarded. Further developments are needed on this issue.

5. Conclusions

A kinetic model based on multigroup radiative transfer equations (MRTE) has been proposed for the vibrational energy evolution in randomly heterogeneous slender structures. It is applicable to the description of high-frequency vibrations of engineering structures subject to unsteady forces of which the spectra typically extend far up the audio-frequency range [2], adopting a time-domain approach. The MRTE have been solved numerically by a discontinuous finite element method and a Monte-Carlo method, among other possible schemes. Both methods have been implemented and tested on examples of

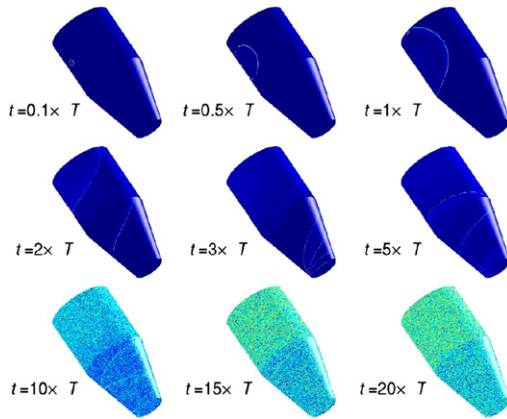


Fig. 13. Same as Fig. 12, with a Rayleigh model of correlation in shell #1.

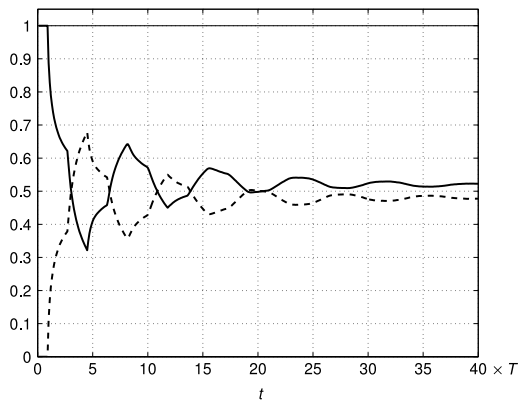


Fig. 14. Evolution of the total vibrational energies in an assembly of two randomly heterogeneous cylindrical shells with $\phi = \frac{\pi}{12}$, $\nu_1 = 0.3$, $\nu_2 = 0.2$, $T = L/c_{T2}$, and $a|\mathbf{k}| = 2$, $\zeta_{11} = \zeta_{22} = 0.1$, $\zeta_{33} = 0.9$, $\zeta_{12} = \zeta_{13} = 0$, $\zeta_{23} = 0.9$ for the scattering cross-sections with a Gaussian model of correlation in shell #1. Thick solid line: shell #1, thick dashed line: shell #2, thin solid line: computed total energy.

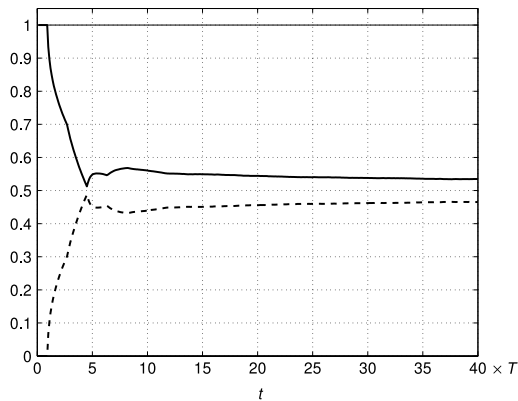


Fig. 15. Same as Fig. 14, with a Rayleigh model of correlation in shell #1.

assemblies of random thick beams or cylindrical shells. Direct numerical simulations illustrate the transfer regime for these systems, and the onset of a diffusive regime at late times. The latter is characterized by energy equilibration rules very much comparable to the assumption of modal equipartition invoked in the statistical energy analysis of structural-acoustics systems [6]. It should be noted in addition that the equipartition limit obtained here in the case of an assembly of two-dimensional thick beams is recovered as well for three-dimensional beam trusses, as demonstrated in [45]. Nevertheless a rigorous mathematical

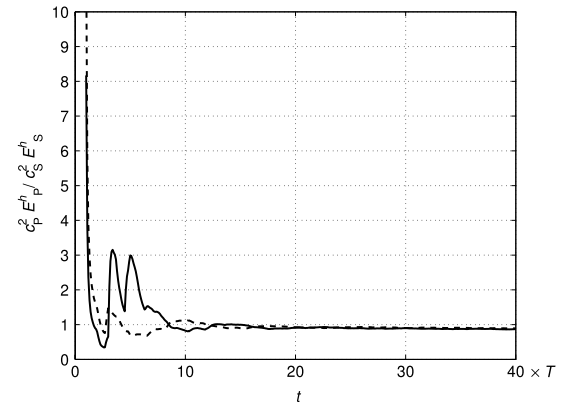


Fig. 16. Evolution of the ratio $c_p^2 E_p^h(t) / c_s^2 E_s^h(t)$ of the total bending/longitudinal and transverse energies in an assembly of two randomly heterogeneous cylindrical shells with $\phi = \frac{\pi}{12}$, $\nu_1 = 0.3$, $\nu_2 = 0.2$, $T = L/c_{T2}$, and $a|\mathbf{k}| = 2$, $\zeta_{11} = \zeta_{22} = 0.1$, $\zeta_{33} = 0.9$, $\zeta_{12} = \zeta_{13} = 0$, $\zeta_{23} = 0.9$ for the scattering cross-sections with a Gaussian model of correlation in shell #1. Solid line: shell #1 ($j = 1$), dashed line: shell #2 ($j = 2$).

model of this diffusive regime for bounded elastic media in a general setting lacks at present. It should be able to predict the equilibration ratios as function of the sub-systems mechanical parameters. Another open issue is the consideration of the glancing set of energy rays – the diffractive and gliding ones – for the part of the power flows possibly trapped by the interfaces and junctions. Both aspects are the subject of ongoing research.

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